

Fraction multiplication ($\times$ fraction) + mixed number processing

Unit 3 · fraction operation (multiplication) · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5A Unit 3 + Modern Education 5A Unit 11

● **High frequency** The score test must be taken every year, accounting for about 12-18% of Paper 1

Prerequisite knowledge: Class 7-8 (Adding common denominators and different denominators), P4 integers Mixed numbers The sub-test is compulsory every year and accounts for about 12-18% of Paper 1.

Prerequisite knowledge: Lesson 7-8 (Addition of common denominators and different denominators), P4 integers $\times$ fractions, HCF and reduction

Our goals: ❶ Cross reduction method ❷ Mixed number $\times$ fraction (conversion to improper fraction method) ❸ Multiplication of three fractions ❹ Fraction multiplication word problems

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

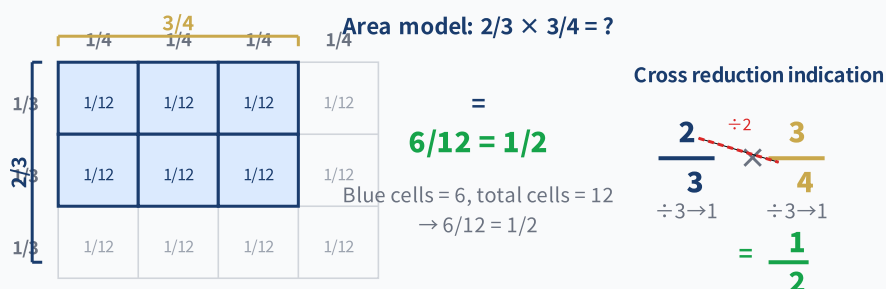
#	Question	Difficulty	Working Space (Show full working)
1	Calculate $3 \times \frac{1}{4} = ?$ (Integer $\times$ Fraction, P4 Basics)	Basic	
2	Calculate $\frac{1}{3} \times 2 = ?$ (fraction $\times$ integer)	Basic	
3	Convert $1\frac{1}{2}$ to an improper fraction. Then find the improper fraction forms of $1\frac{2}{3}$ and $2\frac{1}{4}$. Improper fraction form.	Basic	
4	Reduce $\frac{6}{15}$ to the simplest fraction. (HCF(6,15) = ?)	Basic	
5	Find HCF(6, 8), HCF(4, 10), HCF(9, 12). Why is HCF needed for cross reduction?	Advanced	

II.Core Knowledge + Worked Examples

Knowledge Point 1: Fraction $\times$ Fraction — Cross Reduction Method ● SSPA

Fraction multiplication rules:

- ① Numerator $\times$ numerator, denominator $\times$ denominator: $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$
- ② **Cross reduction (reduce first and then multiply) |||SEP|||: Before multiplication, first combine any numerator with any denominator Use HCF to reduce**
- ③ Advantages: reduce first and then multiply $\rightarrow$ numbers become smaller $\rightarrow$ fewer errors $\rightarrow$ no need for final reduction
- ④ Note: reduction can only be between numerator and denominator |||SEP|||, not numerator and numerator, denominator and denominator! **between numerator and denominator**, cannot numerator with numerator, denominator with denominator!



□ **Example of trap detonation (the most important demonstration in this class)**

calculate: $\frac{2}{3} \times \frac{3}{4} = ?$

✗ **Common mistakes (60% students)**

✓ **Correct solution (cross reduction)**

$$\frac{6}{12} \text{ (not reduced)}$$

Directly multiply and then forget to reduce → Not the simplest answer

$$\frac{1}{2}$$

Cross reduction: numerator 2 and denominator 4 are reduced (HCF=2) → $13 \times 32 = 36 = 12$

 **Tip: "When doing cross-reduction, do it first. The numerator and denominator are about one round. Multiply the rest first and then check. The answer must be the simplest one."**

 **The most frequent error: after multiplying $2/3 \times 3/4 = 6/12$, take it as the answer directly → Forget about reduction! Cross reduction can avoid this error.**

 **The second most frequent error: When approximating a division, the numerator is about the numerator and the denominator is about the denominator → This is wrong! It can only be "crossed" (the numerator is about the denominator).**

Knowledge Point - Worked Examples (must first cross simplify, again multiply)

#	Question	Difficulty	Working Space
Example 1	$\frac{1}{2} \times \frac{2}{5} = ?$ (Hint: Reduction of numerator 2 and denominator 2 → $\frac{1}{5}$)		
Example 2	$\frac{3}{4} \times \frac{2}{9} = ?$ (Hint: Numerator 3 is approximately the same as the denominator 9, Numerator 2 is approximately the same as the denominator 4)		

Knowledge Point 2: Mixed Numbers

Standard procedure for multiplication of mixed fractions:

① **Conversion to improper fractions** |||SEP|||: Mixed fractions → Improper fractions (integer × denominator + numerator) Cross reduction |||SEP|||: Reduction between numerator and denominator

② **Numerator × numerator, denominator × denominator** The answer is converted back

③ : **Improper fraction → Mixed fraction (if necessary) + reduction**

④  **Absolutely prohibited:** Direct multiplication of mixed fractions! Integer × Integer + Fraction × Fraction ≠ Correct answer!

 **Absolutely prohibited:** Multiply mixed fractions directly! Integer × Integer + Fraction × Fraction ≠ Correct answer!

 **Wrong way**

$$1 \text{ and } 1/2 \times 1 \text{ and } 1/3 \\ \neq (1 \times 1) + (1/2 \times 1/3) = 1 \text{ and } 1/6$$

→

 **Correct way**

$$1 \text{ and } 1/2 = 3/2 \rightarrow 3/2 \times 1 \text{ and } 1/3 = 4/3 \\ = 12/6 = 2$$

$$1 \text{ and } 1/2 = 3/2$$

$$\begin{array}{|c|c|} \hline 1 & 1 \\ \hline \end{array} 1$$

$$1 \text{ and } 1/3 = 4/3$$

$$\begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline \end{array} 1$$

$$3/2 \times 4/3 = 12/6 = 2$$

1

Convert Improper Fractions

Mixed fraction → Improper fraction

2

cross reduction

The numerator and denominator are reduced using HCF

3

Multiply

Numerator × Numerator
Denominator × Denominator

4

Reverse + reduce

False → mixed fraction
simplest fraction

□ Trap comparison question

Calculate: $1\frac{1}{2} \times \frac{1}{3} = ?$

✗ **Error: Multiply mixed fractions directly**

$$1 \times 1 + \frac{1}{2} \times \frac{1}{3} = 1\frac{1}{6}$$

Totally wrong! Mixed numbers must be converted to improper fractions first

✓ **Correct: Convert improper fractions first**

$$\frac{1}{2}$$

$$1\frac{1}{2} = \frac{3}{2} \rightarrow \frac{3}{2} \times \frac{1}{3} = \frac{3}{6} = \frac{1}{2}$$

example

Example 3: $1\frac{1}{2} \times \frac{1}{3} = ?$

example

Example 4: $2\frac{1}{3} \times \frac{3}{4} = ?$

Knowledge point 3: Multiplying three fractions ● SSPA Advanced

Rule of continuous multiplication:

- ① Multiply all the numerators and all the denominators
- ② **Comprehensive cross reduction |||SEP|||: Any numerator can be reduced with any denominator (not limited to pairing)**
- ③ Suggestion: Comprehensive reduction first → then multiply → the number becomes super small!
- ④ Example:

(approximate first, then multiply) $\frac{a}{b} \times \frac{c}{d} \times \frac{e}{f} = \frac{a \times c \times e}{b \times d \times f}$ (Make an appointment before taking a ride)

example

Example 5: $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = ?$ (Hint: Numerator 1, 2, 3 intersects with denominator 2, 3, 4 approximately)

example

Example 6: $\frac{2}{3} \times \frac{3}{5} \times \frac{5}{6} = ?$



Formula (continuous multiplication version): "Multiple three brothers in a row, make an appointment together, and then multiply together after the appointment, the answer is so simple."

Knowledge Point 2 + 3 Synchronous practice

#	Question	Difficulty	Working Space
6	$1\frac{1}{4} \times \frac{1}{2} = ?$		
7	$1\frac{2}{3} \times \frac{1}{5} = ?$		
8	$\frac{1}{2} \times \frac{1}{4} \times \frac{4}{5} = ?$		
9	$\frac{1}{3} \times \frac{3}{4} \times \frac{4}{5} = ?$		

III. Lesson layered synchronization practice (total 15 questions)

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
10	$\frac{1}{2} \times \frac{1}{3} = ?$		
11	$\frac{1}{4} \times \frac{1}{2} = ?$		
12	$\frac{2}{3} \times \frac{1}{2} = ?$		
13	$\frac{3}{4} \times \frac{1}{3} = ?$		
14	$\frac{1}{5} \times \frac{5}{6} = ?$		

Advanced layer (total 5 questions, choose do)

#	Question	Difficulty	Working Space
15	$\frac{3}{5} \times \frac{5}{6} = ?$		

16	$\frac{2}{3} \times \frac{3}{8} = ?$		
17	$1\frac{1}{4} \times \frac{1}{3} = ?$		
18	$2\frac{1}{2} \times \frac{1}{5} = ?$		
19	$\frac{1}{2} \times \frac{1}{3} \times \frac{1}{4} = ?$		

 challenge layer (total 5 questions,  choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
20	$1\frac{2}{3} \times \frac{3}{5} = ?$		
21	$2\frac{1}{3} \times 1\frac{1}{2} = ?$		
22	$\frac{3}{4} \times \frac{8}{9} \times \frac{1}{2} = ?$		
23	$1\frac{1}{4} \times \frac{2}{3} \times \frac{3}{5} = ?$		
24	$3\frac{1}{3} \times \frac{3}{5} \times \frac{1}{2} = ?$		

Knowledge point 4: Fraction multiplication word problems (subject to sub-test word questions) SSPA compulsory exam

Four steps to solve the problem:

- ① **Circle keywords**("... of..." = multiplication! "× divided by
- ② **Write a complete answer**(Step points will be deducted if you do not write an answer!)
- ③ **Key words**:"C divided by B" = $A \times$
- ④ **Write a complete answer**(Points for steps will be deducted if you don't write an answer!)

Key words:"A divided by B divided by C" = $A \times \frac{C}{B}$

example

Example 7: There are $\frac{2}{3}$ pieces in a cake, and Xiao Ming eats $\frac{1}{2}$ pieces among them. What fraction of the whole cake did Xiao Ming eat? (Hint: "where" $\frac{1}{2} =$ multiplication)

example

Example 8: A rope is $1\frac{1}{2}$ meter long, use |||SEP|||. How many meters were used? (Tip: Convert mixed fractions to improper fractions first and then multiply them) $\frac{3}{5}$. How many meters were used? (Tip: Convert mixed fractions to improper fractions first and then multiply them)

applicationquestionpractice (all must do, column → simplify → calculate → answer sentence)

#	Question	Difficulty	Working Space
25	A box of biscuits weighs $\frac{2}{3}$ kilograms, and Xiaomei ate SEP . How many kilograms did Xiaomei eat? $\frac{1}{4}$. How many kilograms did Xiaomei eat?		
26	A bottle of juice contains $\frac{3}{4}$ liters. Xiao Ming poured out SEP . How many liters were poured? $\frac{2}{3}$ come out. How many liters were poured?		
27	A rectangle is long $\frac{1}{2}$ meters and its width is long SEP . How many meters wide is it? $\frac{1}{2}$. How many meters wide is it?		
28	The elder brother has $1\frac{1}{2}$ kilogram of candy, and he gives $\frac{1}{3}$ to his younger brother. How many kilograms were sent?		
29	Mom used $\frac{3}{4}$ kilograms of flour to make cakes, of which $\frac{2}{5}$ was used to make cookies. How many kilograms of flour are used to make cookies?		
30	A ribbon is $2\frac{1}{4}$ meters long, and my sister used SEP . How much rice did your sister use? (Answer with mixed fractions) $\frac{2}{3}$. How much rice did your sister use? (Answer with mixed fractions)		

IV. Ultimate challenge area (total 3 questions)

#	Question	Difficulty	Working Space
 1	$\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} = ?$ (Multiplying four fractions - the ultimate in cross reduction!)		
 2	The rectangular area $\frac{2}{3}$ is planted with roses, and the rose part with $\frac{1}{4}$ is a red rose. What fraction of the total area do red roses occupy?		

 3	The granary contains $5\frac{1}{3}$ tons of rice. $\frac{1}{4}$ was used in the first week, and $\frac{1}{3}$ was used in the second week (note: is it "the rest" or "all"?). The title says "I used $\frac{1}{3}$ in the second week", which refers to the total amount of $\frac{1}{3}$ used. How many tons were used in two weeks? How many tons are left?		
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V. Class afterhomework

Basic must do questions (total 5 questions, must write cross simplify step)

#	Question	Difficulty	Working Space
H1	$\frac{1}{2} \times \frac{1}{2} = ?$		
H2	$\frac{2}{3} \times \frac{1}{4} = ?$		
H3	$\frac{1}{3} \times \frac{3}{5} = ?$		
H4	$\frac{3}{4} \times \frac{2}{3} = ?$		
H5	$1\frac{1}{2} \times \frac{1}{4} = ?$		

Advanced choose doquestion (total 3 questions,  choose do)

#	Question	Difficulty	Working Space
H6	$1\frac{2}{3} \times \frac{3}{5} = ?$		
H7	$2\frac{1}{2} \times 1\frac{1}{3} = ?$		
H8	A rectangular garden $\frac{3}{4}$ is planted with flowers, and the flowers $\frac{1}{3}$ are roses. What percentage of the garden are roses?		

VI. The Lessoncorecommon errorssummary

 Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point
 ☐ I can complete  basic questions independently

☐ I can challenge  advanced questions
 ☐ I remember the formula

 Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall
 ☐ Solve  basic questions independently (100% correct)

#	common error	Correct Approach
1	Cross reduction is done : $\frac{2}{3} \times \frac{3}{4} = \frac{6}{12}$ when the answer	First make a cross reduction (the numerator 2 and the denominator 4 are approximately $\rightarrow \frac{1}{3} \times \frac{3}{2} = \frac{3}{6} = \frac{1}{2}$)
2	Multiply mixed numbers directly without converting them into improper fractions	The first step in multiplication with mixed numbers = Convert Improper Fractions! No exceptions!
3	The numerator is about the numerator, the denominator is about the denominator	The reduction can only be between the numerator and the denominator (crossover), the numerator cannot be reduced!
4	The result is unreduced : $\frac{6}{15}$ When the final answer	Finally the HCF reduction must be checked. Cross-reduction first can avoid this problem.
5	Improper fractions are not converted to mixed fractions	When the answer is > 1 , the test is required to be converted into a mixed number
6	When multiplying continuously, only one pair of numerator and denominator will be arranged and then stop SEP : Only one pair of numerator and denominator will be arranged, and no other appointment will be made: Only a pair of numerator and denominator are agreed, and no other appointments are made.	When multiplying, all numerators and all denominators can be cross-contracted! Total simplification.
7	Missing answers to application questions/missing units	Must write "Answer:..." + unit, otherwise step points will be deducted

Lam Fung Academy • LF Academy • We don't teach math. We teach trap avoidance.

 Related topics: L07 Fractions with different denominators • L10 Fraction word problems • L22 Fraction division • Related topics: L05 Multiplication knowledge • L06 Two-digit numbers multiplied by one digit • L07 Multiplication word problems